INCH-POUND

MIL-D-51559(EA)
10 October 1991

MILITARY SPECIFICATION

DECONTAMINATING APPARATUS: PORTABLE, 14 LITER, M13 (PRACTICE)

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of assembled practice decontaminating apparatus (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-636 - Boxes, Shipping Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4230

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PPP-F-320 - Fiberboard: Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes.

STANDARDS

MILITARY

MIL-STD-1168 - Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-51-608	-	Decontaminating Apparatus: Portable, 14 Liter, M13 (Practice)
5-51-609	-	Packaging and Marking (Practice)
5-51-610	-	Palletization and Marking (Practice)
5-51-611	-	Packing and Marking (Practice)

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-IOT, Rock Island, IL 61299-7210.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials cited on Drawing 5-51-608 and on the subsidiary drawings shall conform to the specifications and drawings listed thereon.

3.2 Components. All components of the practice apparatus shall conform to the specifications and drawings listed on Drawing 5-51-608 and subsidiary drawings.

3.3 Manufacture and assembly. The practice apparatus shall be manufactured and assembled in accordance with Drawing 5-51-608.

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3.4 Spray characteristics. Using the Government-furnished DS2-filled fluid container (see 3.6) in lieu of the empty plastic container (Drawing 5-51-598), at room conditions the apparatus shall be capable of being primed with no more than 10 strokes of the pump piston. With the wand held horizontally a minimum of 4 feet above the outlet of the DS2-filled fluid container, spray shall be projected a horizontal distance of at least 3 feet and no leakage shall occur when the pump is activated and tested in accordance with 4.4.6.1.

3.5 First article. Unless otherwise specified in the contract, a sample of practice apparatus shall be subjected to the first article inspection specified in 4.3 prior to the start of regular production and whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material, drawings, specifications, or source of supply as to significantly affect product uniformity as determined by the Government.

3.6 Government-furnished equipment (see 6.5). The Government will furnish DS2-filled fluid containers (Drawing 5-51-575) to the contractor to conduct the spray characteristics test (4.4.6.1).

3.7 Workmanship. The practice apparatus shall be free from cracks, dents, sharp edges, unpainted surfaces, rust, and foreign matter such as dirt, grease, and oily or viscous materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.3.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Inspections to be performed. As determined by the Government, the first article sample items shall be examined and tested in accordance with this specification using either the first article procedures in 4.3.3 or other first article procedures which will be provided by the Government. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.3.2 Sample. The first article sample shall consist of 100 practice apparatus manufactured and packaged using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.3 Inspection procedure (see 6.4).

4.3.3.1 For examination and test. Twenty practice apparatus shall be taken at random from the first article sample and subjected to the spray characteristics test specified in 4.4.6.1. Twenty decontaminating apparatus and the packaging thereof shall be taken at

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random from the first article sample and examined for all of the remaining characteristics in the classification of characteristics using the inspection methods specified therein.

4.3.4 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of practice apparatus shall consist of those like items produced at the same place utilizing the same batches of materials, lot of components, process runs, fabrication techniques, assembly methods, tools, equipment and facilities but shall not exceed one month's production. Each lot shall be controlled and identified in accordance with MIL-STD-1168. No more than one interfix lot number of filled fluid containers shall be contained in any one lot of practice apparatus.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.5 and, when specified, table I. Samples shall be selected at random.

4.4.3 Inspection procedure (see 6.4). Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

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TABLE I. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.4.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(a)	Decontaminating apparatus: portable, 14 liter, M13 (Practice)		5-51-608
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Width	Table I, Level VII	3.2	CE
102	Height	Table I, Level VII	3.2	CE
103	Length	Table I, Level VII	3.2	CE
104	Spray characteristics	4; Acc 0, Rej 1	3.4	4.4.6.1
105	Specified components correctly assembled	Table I, Level VII	3.2 and 3.3	VI
Minor				
201	Labels evident and correctly assembled	Table I, Level IX	3.2	VI
202	Workmanship	Table I, Level VII	3.7	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(b)	Packaging - Packing - Palletization		5-51-609, 5-51-610 5-51-611
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Item completely clean and dry prior to unit packing	Table I, Level VII	5.1	VI
102	Filler pads evident and correct	Table I, Level VII	5.1	VI and CE
103	Cushioning evident and correct	Table I, Level VII	5.1	VI and CE
104	Unit pack container evident and correct	Table I, Level VII	5.1	VI and CE
105	Unit pack closure evident and correct	Table I, Level VII	5.1	VI and CE
106	Unit pack container marking evident, correct and undamaged	Table I, Level VII	5.3	VI and CE
107	Shipping container and closure correct and undamaged	Table I, Level VII Table I, Level VII	 5.2	VI and CE VI and CE
108	Shipping container marking evident, correct and legible	Table I, Level VII	5.3	VI and CE
Minor				
201	Pallet correct and undamaged	Table I, Level IX	5.4	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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4.4.6 Tests (see 6.4). Unless otherwise specified, tests shall be performed at room temperature, 20° to 30°C (68° to 86°F), as follows:

4.4.6.1 Spray characteristics.

(a) Assemble the unit in accordance with the operating instructions (Drawing 5-51-571) using the Government-furnished DS2-filled fluid container (see 3.6) in lieu of the empty plastic container.

(b) Condition container to room temperature.

(c) With the valve open, operate the pump until fluid flows from the wand end. Record the number of strokes required.

(d) Hold the wand in specified position, operate pump and determine if pump projects a horizontal spray the required distance.

(e) Continue to operate for a minimum of five additional strokes and observe entire assembly for leakage.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be levels A or C as specified (see 6.2).

5.1.1 Level A. The practice apparatus shall be preserved and packaged in accordance with Drawing 5-51-609.

5.1.2 Level C. The practice apparatus shall be unit packed in accordance with Drawing 5-51-609, except the RSC fiberboard box and pads shall conform to type CF, class domestic, variety SW, grade 175 of PPP-B-636 and PPP-F-320, respectively. The box shall be closed in accordance with the appendix of PPP-B-636 for domestic class boxes.

5.2 Packing. Packing shall be levels A, B or C as specified (see 6.2).

5.2.1 Level A. The practice apparatus preserved and packaged as specified in 5.1.1 shall be packed in accordance with Drawing 5-51-611.

5.2.2 Level B. The practice apparatus preserved and packaged as specified in 5.1.1 shall be packed in accordance with Drawing 5-51-611, except that the shipping container shall conform to class 1, style 1 of PPP-B-585.

5.2.3 Level C. The practice apparatus, unit packed as specified in 5.1.1 or 5.1.2, shall require no further protection.

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5.3 Marking. In addition to any special marking required by the contract or order, unit and shipping containers shall be marked in accordance with Drawings 5-51-609 and 5-51-611, as applicable. When required, the palletization unit load shall be marked in accordance with Drawing 5-51-610.

5.4 Palletized unit loads. When required, practice apparatus packed in accordance with 5.2.1 and 5.2.2 shall be palletized on standard type, 4-way, 40 by 48 inch pallets in accordance with Drawing 5-51-610.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M13 Decontaminating Apparatus (Practice) is a man-portable device intended for use by personnel who require practice in the operation of the M13 Decontaminating Apparatus (DAP). The practice apparatus is equipped with an empty plastic container which is to be filled with water or other DS2 simulant in lieu of the DS2-filled fluid container of the M13 DAP.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
(c) Level of preservation, packaging, packing, palletization required and shelf life code (see section 5).
(d) First article:

(1) First Article Inspection and Report requirements.
(2) Delivery schedule for First Article Report.
(3) Destination and delivery schedule of First Article Samples when testing is performed by the Government.
(4) First Article Test report approval time.

(e) Production

(1) Destination and delivery schedule requirements.
(2) Production and inspection report requirements.

6.3 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for

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evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Sampling and testing precautions. DS2 is a combustible liquid (Flash Point 168°F) and should not be stored near open flames or heat sources. DS2 is classified as a corrosive liquid as it contains sodium hydroxide and diethylenetriamine which are caustic in nature. DS2 is a toxic chemical with an ammonia-like odor and should be handled in a well ventilated environment and with appropriate protective clothing and equipment (i.e. corrosive resistant gloves, boots, apron, an industrial faceshield or splash-proof chemical goggles, and full facepiece respirator with amine cartridge). Use copious amounts of water for eye or skin contact. If DS2 is ingested, administer as much milk or water as possible, but do not induce vomiting. Spills should be neutralized with sodium bisulfate, absorbed with vermiculite and containerized as quickly as possible. Fires involving DS2 can be extinguished with CO_2, water spray, or alcohol foam.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.6.

6.6 Subject term (key word) listing.

Chemical agents
Decontamination
Decontamination equipment
DS2

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

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